Algorithm for file updates in Python

Project description

Update a file that identifies employees who can access restricted content based on the employees ip address, create an algorithm that uses Python code to automate the task of checking if the allow list contains any ip addresses that are identified on the restricted list and remove those ip addresses from the allow list.

Open the file that contains the allow list

with open("import_file","r") as file:

Using a with statement open the file and store it as a variable in order to read the contents.

Read the file contents

ip_addresses=file.read()

To read the file and store the contents for reading use the .read() syntax.

Convert the string into a list

ip_addresses = ip_addresses.split(",")

To convert the string into a list to remove individual ip addresses use the .split(“,”) syntax.

Iterate through the remove list

if element in remove_list:

To iterate throught the remove list an if statement must be written to include the name of the variable to be removed and the list from which to remove it.

Remove IP addresses that are on the remove list

ip_addresses.remove(element)

A conditional must be written, evaluate if the loop variable element is part of the ip_addresses list. Then within that conditional, apply the .remove() method to the ip_addresses list and remove the IP addresses identified in the loop variable element.

Update the file with the revised list of IP addresses

```
 file.write(ip_addresses)
```

The .write() function allows us to update the file with the revised list of ip addresses. Applying the .remove() method in this way is possible because there are no duplicates in the ip_addresses list.

Summary

Utilizing the Python syntax allows us to automate cybersecurity tasks such as reading, updating, and appending lists easily and also saves a lot of time and risk for human error while scanning through lists of ip addresses (for example) using Python.